Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

FIIT-XXXX-XXXXX

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# Deep Learning Methods for Perception and Prediction in Autonomous Systems

Master's Thesis

Thesis Supervisor: Ing. Matej Halinkovič, PhD.

May 20xx

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Master's Thesis

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|---|---|
| Study Programme: | Intelligent Software Systems |
| Study Field: | Computer Science |
| Training Workplace: | Bratislava, Slovakia |
| Thesis Supervisor: | Ing. Matej Halinkovič, PhD. |

May 20xx

Čestne vyhlasujem, že som túto prácu vypracoval samostatne, na základe konzultácií a s použitím uvedenej literatúry.

I declare in my honor that I have prepared this work independently, on the basis of consultations and using the mentioned literature.

In Bratislava, May 20xx

Bc. Richard Čerňanský

## Annotation

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies
Study Programme: Intelligent Software Systems
Study Field: Computer Science

Author: Bc. Richard Čerňanský
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**Keywords:** 3-5 keywords

## Anotácia

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**Oblasť problematiky:** 3-5 kľúčových slov

BP2, DP3: sem vložte finalne a podpisane zadanie: PDF z AISu

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# Chapter 1

# Introduction

In recent years, deep learning and computer vision methods have made significant advancements in the analysis of dynamic scenes and their applications in autonomous systems. The development of fields such as autonomous vehicles and robotics depends on solving tasks like object detection, object tracking, segmentation, and trajectory prediction. Specifically, autonomous vehicles (AVs) revolutionize transportantion by saving drivers' time, improving safety and possibly mitigating congestion on the roads. Current autonomous driving systems consist of multiple sequential tasks such as perception, tracking, prediction, planning and control [1]. Perception task is responsible for detecting and understanding the objects within the environment including their categorization, location and size estimation. The tracking objective uses the detections from perception to find the relationships between objects across the frames in scene. The prediction modules then process this tracking information in order to forecast future positions. The predicted positions enter the planning stage that generates the most feasible and optimal path in terms of safety, energy consuption and following traffic rules. Last but not least, controlling task then prepares actuator commands which execute the planned motion.

## 1.1 Motivation

Real traffic is highly dynamic and unpredictable: other drivers may behave erratically, and sensors provide noisy or incomplete data. Diverse maneuvers, complex interactions, and inherent errors in sensory information make prediction challenging, yet solving these challenges is essential for having trustworthy AVs. In addition, autonomous systems operate under strict constraints on latency and reliability [2]. Predictions must be computed in real time on embedded hardware and remain accurate over long hori-

zons to be useful for driving in real-life scenarios. Industry recognizes the need for fast execution, uncertainty handling, and generalization in prediction models to ensure robust operation in varied conditions [3, 4, 5]. Reliable trajectory prediction is therefore critical for safe autonomous driving and motivates intensive research into methods that can forecast the multi-agent future with practical efficiency. For these reasons, this thesis focuses on trajectory prediction, exploring how current state-of-the-art models could be modified or extended to achieve better accuracy and efficiency.

## 1.2 Problem Formulation for Trajectory Prediction

Trajectory prediction is the task of forecasting the future motion of the traffic participants that surround the ego vehicle. Its input is not a trajectory in the purely geometric sense, but rather the past states of these participants as they are captured in the environmental information collected by sensors, such as camera images, LiDAR point clouds, radar returns or HD maps providing lane geometry and traffic-rule priors. However, in this thesis only camera and LiDAR is employed. A model that relies on a single source is unimodal, whereas a model that exploits several of them at once is considered multi-modal. In the latter case the measurements are heterogeneous in dimensionality, sampling rate and coordinate frame, so the features extracted from the individual sources have to be fused into a unified representation of the scene before any forecasting takes place [6]. The output is the forecasted trajectory, that is, a sequence of future states of the observed agents over a fixed prediction horizon.

Formally, trajectory prediction can be stated as the problem of constructing a mapping $\Phi$ from the observed evidence $\mathcal{I}$ about the scene to the future states $\mathcal{O}$ of the traffic participants, from the current time up to a prediction horizon $T$ [7]:

$$\begin{aligned} &\Phi : \mathcal{I} \rightarrow \mathcal{O}, \\ &\mathcal{O} = \left\{ S^0, S^1, \ldots, S^T \right\}, \end{aligned} \tag{1.1}$$

where $S^0$ is the current state and $S^1$ to $S^T$ are the states at the individual future time steps, each of them collecting the states of all $N$ observed participants. Depending on the available modalities and on the downstream task, a single state can take different forms, for example

$$S^t = \left\{ x^t, y^t \right\} \quad \text{or} \quad S^t = \left\{ x^t, y^t, l^t, w^t, \theta^t \right\}, \tag{1.2}$$

where $x^t$ and $y^t$ are the coordinates of the centre point of the object, $l^t$ and $w^t$ its length and width, and $\theta^t$ its orientation [7]. In trajectory prediction the state is usually reduced to the planar position, so that $\mathcal{O}$ corresponds to the set of forecasted trajectories of the

participants in the scene. Since the future of a traffic scene is inherently uncertain, the mapping in (1.1) is in practice often extended to output several plausible trajectories together with their confidences, allowing the planning stage to reason over the alternatives instead of committing to a single hypothesis [2]. The following subsections summarise the main challenges that this formulation raises.

### 1.2.1 Data Fusion

The information describing the scene comes from sources that are complementary but incompatible. Cameras provide dense semantic detail without depth and degrade at night, in glare or in poor lighting conditions, LiDAR provides accurate geometry but suffers in rain etc. These sources can be combined at the level of raw measurements (early), of learned features (intermediate) or of the individual predictions (late). Since every added modality costs latency and introduces noise the network is prone to overloading with parameters or can overspecialize on the most informative source and lose accuracy once the weather or illumination degrades it [6].

### 1.2.2 Dynamic Environments

Traffic is nonstationary, as density, weather, illumination and road layout change, so that the conditions met during deployment need not resemble those seen during development. The future is moreover not unique, since the same observed history is usually compatible with several maneuvers, and this aleatoric uncertainty originated in the erratic decisions of other drivers does not diminish with additional data. Therefore the task requires a set of socially compliant trajectories produced in time for the planning stage [6].

### 1.2.3 Interaction Modeling

The motions of traffic participants are not independent and traffic signals constrain all of them, so that the joint future does not split into separate per-agent distributions. Capturing the agent to agent and the agent to map relations requires representations able to operate on a variable and unordered number of entities, which is why structured formulations such as graphs and attention mechanisms prevail. Where these relations are modelled poorly, forecasts that appear plausible for each agent in isolation become inconsistent at the level of the scene comprising of multiple entities [2].

### 1.2.4 Real-World Constraints

Prediction is one stage of a control loop closing at a fixed rate, which leaves it a budget of tens of milliseconds on embedded hardware limited in memory and power. Its cost

grows with the number of participants and becomes quadratic once their interactions are modelled, so the least favourable case is dense traffic, precisely where prediction matters most. Model capacity therefore has to be weighed against computational efficiency, as accuracy alone does not make a model deployable [4, 2].

## 1.3 Thesis Outline

# Chapter 2

# Trajectory Prediction: Background and State of the Art

TODO: concrete header for the given thesis's theme!

## 2.1 Taxonomy of Trajectory Prediction Methods

In the literature, there are numerous different prediction methods, and several criteria enable their systematic classification, such as the modality and the representation of the input, the form of the produced output, or the paradigm by which prediction is integrated with perception. Here the modeling methods are classified according to the categorization presented in [6].

### 2.1.1 Physics-Based Methods

These rely on kinematic or dynamic models of motion, using physical equations to extrapolate future positions and not accounting for uncertainty. They are the oldest family in the field, resting on estimators formulated between the 1960s and the 2000s. Examples include constant-velocity and constant-acceleration models (Single-Trajectory simulation), applied to vehicle path prediction and to collision warning in the early 2000s [8, 9]. The Kalman filter, formulated in 1960 [10], addresses the uncertainty limitation of the previous models by modeling the vehicle's state as a Gaussian distribution. It predicts trajectories as a two step process of filtering (denoising current state from noisy measurements) and forecasting. Monte Carlo methods, and in particular the particle filter introduced in 1993 [11], drop the linearity and Gaussianity assumptions of the Kalman filter by approximating the state distribution with a set of weighted

samples propagated through the motion model, which permits arbitrarily shaped and multimodal distributions at the cost of a higher computational load. Physics-based approaches are simple and fast, but they struggle with complex maneuvers or sudden behavior changes since they lack contextual understanding. Some variants combine multiple motion models to handle different maneuvers (Multiple-Model methods), a family surveyed comprehensively in 2005 [12].

### 2.1.2 Learning-based methods

Unlike physics-based, learning-based methods learn the motion dynamics from data. Here, three groups of approaches are mentioned.

#### 2.1.2.1 Classical Machine Learning Methods

These methods learn motion patterns from data using statistical models, but without deep neural networks. They were the dominant data-driven approach before deep learning took over, and they continued to appear in hybrid form well after it. The earliest representatives used here are Gaussian processs (GPs) applied to human motion in 2008 [13] and Gaussian mixture models (GMMs) used for probabilistic vehicle trajectory prediction in 2012 [14], followed by hidden Markov models (HMMs) for maneuver classification in 2018 [15] and by dynamic Bayesian networks (DBNs) combined with sequence learning in 2020 [16]. They can capture uncertainty to a degree (e.g., an HMM can model discrete maneuver states, GPs provide a distribution over trajectories). However, their capacity to model very complex, high-dimensional scenarios is limited compared to modern deep nets.

#### 2.1.2.2 Reinforcement Learning Methods

Reinforcement learning (RL) methods treat prediction as a sequential decision problem. Here, an agent learns a policy (via reward feedback) to mimic human driving behavior, effectively planning likely trajectories. Since the reward that drives human driving is not known, inverse reinforcement learning (IRL) recovers it from expert demonstrations and uses it to regularise the predictor [17], while imitation learning (IL) skips the reward altogether and extracts the policy directly from the observed trajectories [18], both lines reaching trajectory prediction around 2020. RL-based predictors can naturally handle multi-agent interactions by modeling how agents might react to each other, and can ensure physically feasible outputs through learned policy constraints. However, RL methods can be data-hungry, harder to train and stabilize, and historically have been more common in planning than in pure trajectory forecasting. They are an active research area for capturing game-theoretic interactions and long-horizon planning in prediction.

#### 2.1.2.3 Deep Learning Methods

Various Deep Learning methods perform successfully in learning motion dynamics and now dominate trajectory prediction. The field moved through recurrent architectures around 2018, convolutional and graph-based ones between 2019 and 2021, and attention-based ones from 2021 onwards, each family addressing a limitation of the previous one. Their strengths and limitations differ mainly according to their architectural design.

**Recurrent Neural Networks**

Recurrent neural networks (RNNs) and long short-term memory (LSTM) capture temporal history via hidden states updated across timesteps. They are able to effectively capture temporal patterns for trajectory prediction task. Historically, single recurrent models were used to map past trajectories to future paths [19], however recent works facilitate groups or stacks of multiple models [20] or employ them for one of the subtasks like detection [21]. They converge reliably and can be trained autoregressively, in the same mode in which they are later run at inference, but they lack spatial reasoning and struggle with long-range temporal dependencies [22].

**Convolutional Neural Networks**

Convolutional neural networks (CNNs) have already succeeded in many tasks, most notably in computer vision [23]. They suit trajectory prediction because trajectories possess a strong spatio-temporal continuity, and because they operate on any input that can be arranged as a grid, which makes them applicable to the camera, the light detection and ranging (LiDAR) and the high-definition map (HD map) modalities alike. In every case they are responsible for feature extraction, capturing the positions of the objects and the environmental context around them. On camera images a convolutional backbone produces per-view features that are lifted into the bird's-eye view (BEV) plane through a predicted depth distribution [24]. On LiDAR point clouds the points are grouped into vertical pillars and flattened into a pseudo-image [25], from which a sparse convolutional backbone forms the BEV feature map [26]. HD maps are rasterised into additional channels of that same grid, so that lane geometry and traffic rules reach the network through the same convolutions as the sensor data [27].

Temporal convolutional networks (TCNs) are able to model temporal dependencies through the causal convolution layers that preserve the temporal order [28]. They extract spatial features from image-like inputs such as BEV or rasterised maps efficiently and enable for fast inference, although the fixed grid on which they operate represents the individual agents only implicitly and is more expensive than a vectorised encoding of the same scene [22].

#### Graph Neural Networks

Graph neural networks (GNNs) represent the scene as a graph whose nodes are the agents and map elements and whose edges encode the relations between them, which makes them a natural fit for modelling spatial and temporal interactions [29]. They excel in dense interaction scenarios by modelling agent-to-agent relations explicitly, but they add unnecessary complexity in sparse environments such as highway driving, and their dynamic-graph variants rely on consistent object identities and therefore inherit the errors of the tracker [6].

#### Transformers

Transformers replace recurrence with self-attention, which relates every agent and every map element to all the others in a single step and thus captures long-range dependencies without having to propagate information through a hidden state [30]. Since attending over all agents, time steps and map elements at once is expensive, practical models factorise the attention along these axes and decode the future with a set of learned queries that produce several trajectories in parallel [4]. Their capacity for long-term dependencies makes them particularly promising for long-horizon prediction, but they are highly sensitive to training stability and demand extensive tuning and warm-up [6]. Transformer systems pair a strong scene encoder with multi-modal decoders and explicit interaction and map context [30, 31].

## 2.2 Prediction Paradigms

In this section, the prediction paradigms are discussed. The prediction paradigm is an important characteristic of a model, as it determines how the driving stack is decomposed into modules, what is optimised during training, and what information is passed from perception to prediction. The advantages and the limitations of a given model largely follow from this choice.

### 2.2.1 Modular (Detect-Track-Predict)

This approach has been of the most common so far implemented in the literature. The paradigm consists of sequential steps of detection, tracking and prediction, which were independently under an extensive research. First, the sensor data is captured and detector localizes the objects present in the scene, usually as a set of bounding boxes with the corresponding class labels, either from LiDAR point clouds [32] or from camera images [33]. The tracking module then associates these detections across consecutive frames [10], opening a tracklet for every newly observed object and preserving its identity as the scene evolves. Finally, the prediction module consumes the active tracklets and fore-

casts the future positions of the corresponding agents [6], commonly by encoding them together with the map in a vectorised form [34, 35]. Even though this sequential decomposition is intuitive and each module can be developed and debugged separately, the stages are optimised in isolation which causes errors to propagate across modules and results in suboptimal performance due to the lack of end-to-end optimization [7].

TODO: add picture

### 2.2.2 Joint Learning (End-to-End Perception & Prediction)

Joint learning methods treat detection and motion forecasting as a single optimization problem instead of two independent stages. These unified optimization goals help to mitigate the error accumulation in the intermediate steps. Interest in the paradigm has grown steadily since the first such models appeared in 2018 [36], to the point that a recent survey dedicated to end-to-end perception and prediction already reviews more than a hundred works [7]. Such models are trained with a compound loss over both tasks. They differ in how tightly the stages are coupled, ranging from a shared LiDAR backbone co-optimized through a combined loss [36], through BEV features reused by both tasks [37], to full gradient flow from the prediction loss back through the detector [38]. Query-based models take this integration further by representing every agent as a learnable query that is iteratively refined while attending to scene and map features, so that detection, tracking and forecasting are resolved together [39, 40]. The price of this coupling is that task boundaries blur, which makes poor performance hard to attribute to a specific component, and the iterative refinement of queries adds computational overhead that complicates real-time deployment in dense scenes [6].

TODO: add picture

TODO: add tree of the methods

## 2.3 Datasets

Research in trajectory prediction has been accelerated by large-scale autonomous driving datasets. They differ in the sensors they record, in the length and the annotation rate of their sequences, and in whether their annotations support perception, forecasting, or both.

- **KITTI (2012)** [41]: the early autonomous vehicle (AV) benchmark with stereo cameras, LiDAR and global positioning system (GPS)/inertial measurement unit (IMU), widely used for detection, tracking and simultaneous localization and mapping (SLAM). With roughly 150 short sequences it supports only short-horizon forecasting and contains little social interaction.

- **ApolloScape (2018)** [42]: Chinese urban driving data covering detection, segmentation, tracking and mapping. It is useful for perception and localization, but it is less standardized for forecasting than nuScenes or Argoverse.
- **Lyft Level 5 (2019)** [43]: a large urban dataset with synchronized LiDAR and cameras and with HD maps, similar in format to nuScenes and used mostly for 3D detection and tracking.
- **nuScenes (2020)** [44]: 1,000 twenty-second scenes recorded with 6 cameras, 5 radars and one LiDAR covering 360°, annotated at 2 Hz with 3D boxes, HD maps and object identities that persist across frames. The official split is 700 training and 150 validation scenes, and the prediction challenge adds future trajectories over a 6 s horizon on exactly these sequences.
- **Argoverse 2 (2021)** [45]: the successor of the benchmark that popularized map-based prediction and established ADE, FDE and MR as the reported metrics [46]. It is published as several distinct collections, among them a motion forecasting set of 250k scenarios with per-scenario HD maps and a separate sensor set of 1,000 annotated sequences with seven ring cameras, two stereo cameras and LiDAR. The two collections contain different recordings.
- **Waymo Open Motion (2021)** [3]: more than 100k twenty-second scenarios sampled at 10 Hz, with HD maps and a large share of interactive situations. It introduced mAP for multi-modal predictions. Later releases of the dataset extend these scenarios with LiDAR point clouds [47] and with camera information, but both cover only the first second of each nine-second window, and the camera data is distributed as learned image tokens rather than as raw images.

NuScenes is used for every experiment reported in this thesis. A jointly trained perception and prediction model needs synchronized raw LiDAR and camera data, 3D boxes, object identities that persist across frames so that tracking can be supervised, and future trajectories, all annotated on the same sequences. Several of the datasets above hold these annotations apart. NuScenes defines an official prediction benchmark on exactly the sequences it annotates, and it is the dataset on which the end-to-end models discussed in Section 2.2 report their results, which makes the numbers obtained here directly comparable with them.

TODO: add table with datasets if training on multiple datasets, and also edit metrics if needed

## 2.4 Evaluation Metrics

A jointly trained model has to be judged on every task it solves, so three families of metrics are reported. The detection metrics measure how well the objects present in

the scene are identified, the tracking metrics measure how reliably their identities are maintained over time, and the trajectory prediction metrics measure how well their future motion is anticipated. All three families are built on a notion of a correct match between a prediction and a ground-truth annotation, but they establish that match in different ways.

### 2.4.1 Detection Metrics

The protocol used throughout this thesis is the one defined with nuScenes [44] and summarized in [48]. A predicted box is matched to a ground-truth box of the same class when the distance between their centres on the ground plane is below a threshold $d$, which replaces the intersection over union criterion common in image-based detection and avoids penalising small objects disproportionately. Matching is greedy over predictions sorted by decreasing confidence, and every ground-truth box can be matched at most once. The outcome of this assignment defines the counts:

- a *true positive* is a prediction matched to a ground-truth box within threshold $d$,
- a *false positive* is a prediction that finds no such match, either because nothing is there or because the object was already claimed by a higher scoring prediction,
- a *true negative* is a correctly reported negative, absence of any detection.
- a *false negative* is a ground-truth box left without any matching prediction.

Based on these, the following performance metrics are defined:

1. **Precision and recall** express how many of the predictions are correct and how many of the annotated objects are found,

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \tag{2.1}$$

   Sweeping the confidence score threshold traces a precision-recall curve $p(r)$.
2. **Average Precision (AP)** summarises that curve for a given class as the area under its interpolated form, so that fluctuations of $p(r)$ at individual operating points do not influence the result [48],

$$AP = \int_0^1 \max\left\{ p(r) : r \geqslant r_0 \right\} dr_0, \tag{2.2}$$

   where $p(r)$ is the precision at recall $r$. The nuScenes protocol evaluates this over the operating points that remain once those with a recall or a precision below 0.1 have been discarded as dominated by noise, and normalises the result to the unit interval [44].
3. **Mean Average Precision (mAP)** averages AP over the set of classes $C$ and over a set of matching thresholds $\mathcal{T}$, so that a method is not rewarded for being accurate

only at one tolerance,

$$\mathrm{mAP} = \frac{1}{|C|\,|\mathcal{T}|} \sum_{c \in C} \sum_{t \in \mathcal{T}} \mathrm{AP}_{c,t}, \qquad \mathcal{T} = \{0.5,\ 1,\ 2,\ 4\}\,\mathrm{m}. \tag{2.3}$$

4. **True positive error metrics** describe how good the matched boxes actually are, which mAP alone does not capture because a match within $d$ counts fully regardless of its quality. They are computed over the matches obtained at $d = 2\,\mathrm{m}$ and averaged over classes. They comprise the **translation error ATE** as the centre distance in metres, the **scale error ASE** as $1 - \mathrm{IoU}$ after aligning translation and orientation, the **orientation error AOE** as the yaw difference in radians, the **velocity error AVE** in metres per second, and the **attribute error AAE** as one minus the attribute classification accuracy.
5. **NuScenes Detection Score (NDS)** combines the two into a single number, weighting the detection quality against the five error terms,

$$\mathrm{NDS} = \frac{1}{10}\left[5\,\mathrm{mAP} + \sum_{\mathrm{mTP} \in \mathbb{TP}} \big(1 - \min(1,\ \mathrm{mTP})\big)\right], \tag{2.4}$$

where $\mathbb{TP}$ is the set of the five averaged error metrics. Each error is **clipped at one**, so an arbitrarily bad error cannot drive the score below the contribution of mAP alone.

### 2.4.2 Tracking Metrics

Tracking is as well evaluated with the protocol of the nuScenes tracking benchmark [44]. Predictions are matched to ground-truth objects frame by frame with the same centre distance criterion as in detection, but the assignment now also has to be consistent in time, because a tracker is required to keep the same identity attached to the same object. The matching therefore proceeds in two stages: a pairing established in an earlier frame is carried forward whenever both partners are still present and within the allowed distance, and only the objects and tracks left over are assigned by Hungarian matching on the centre distance. What is understood as a *true positive* therefore changes:

- a *true positive* is a ground-truth object matched within the allowed distance that either carried no track before or is matched to the track it already carried,
- a *false positive* is a track that is matched to no ground-truth object, so the tracker is following something that does not exist,
- a *false negative* is a ground-truth object that no track reached in this frame,
- a *true negative* is a correctly reported negative, absence of any detection.
- an *identity switch (IDS)* is a ground-truth object matched to a track whose identity

differs from the one it carried before. A track that is lost and later recovered under a new identity therefore produces an IDS even though every individual box may be accurate.

It is also important to state that the lifetime of a trajectory is not simply the set of frames in which it was annotated. The benchmark first discards annotations that fall outside the per-class evaluation range or that contain no LiDAR returns at all, and then linearly interpolates each trajectory across any remaining interior gap, so that a fully occluded object is restored. A trajectory's lifetime is therefore the uninterrupted span between its first and last surviving observation, and a tracker is expected to carry an identity through occlusions that the sensor itself did not observe. The tracking metrics are as follows:

1. **Multi-Object Tracking Accuracy (MOTA)** aggregates all the ways a tracker can fail into a single number, counting every false positive, every missed object and every identity switch against the number of annotated object instances GT summed over all frames,
$$\mathrm{MOTA} = 1 - \frac{\mathrm{FP} + \mathrm{FN} + \mathrm{IDS}}{\mathrm{GT}}. \tag{2.5}$$
The measure is not bounded from below, as a tracker that reports far more objects than exist can drive it to an arbitrarily negative value.
2. **Multi-Object Tracking Precision (MOTP)** describes how precisely the matched objects are localised, as the mean centre distance $d_i$ over the set of true positives,
$$\mathrm{MOTP} = \frac{1}{|\mathrm{TP}|} \sum_{i \in \mathrm{TP}} d_i. \tag{2.6}$$
3. **Average Multi-Object Tracking Accuracy (AMOTA) and Average Multi-Object Tracking Precision (AMOTP)** remove the dependence on the confidence threshold in the same way AP does for detection. The tracking score is swept, and the two measures above are averaged over a set $\mathcal{R}$ of forty recall levels, from which those below 0.1 are discarded as dominated by noise,
$$\mathrm{AMOTA} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \mathrm{MOTAR}_r, \tag{2.7}$$
where $\mathrm{MOTAR}_r$ is the value of MOTA at recall $r$, normalised so that it occupies the unit interval at every recall level and is therefore comparable across them. AMOTA is the primary measure of the benchmark.
4. **Supporting counts** describe the failure modes that the aggregate scores hide. Unlike MOTA, whose unit is a single object instance in a single frame, these are evaluated along the whole lifetime of one annotated object and each trajectory is

counted once. A trajectory is reported as **mostly tracked (MT)** when it is followed for at least 80% of its lifetime and as **mostly lost (ML)** when it is followed for at most 20%. The remaining trajectories are **partially tracked (PT)**, so that the three categories partition the annotated trajectories,

$$N_{traj} = MT + PT + ML, \tag{2.8}$$

where each count refers to whole trajectories and $N_{traj}$ are all annotated trajectories that survive the evaluation filters.
A **fragmentation** is counted whenever a correctly followed trajectory is interrupted and later resumed. The **track initialisation duration** expresses how long it takes before an object is picked up for the first time, and the **longest gap duration** how long it remains lost in the worst interruption.

These supporting counts matter for a model whose tracks are gated by a confidence threshold. A detector whose individual boxes are accurate but whose scores fluctuate around that threshold loses and reacquires its objects, which leaves AP almost unaffected while raising the number of fragmentations and identity switches and therefore lowering AMOTA.

### 2.4.3 Trajectory Prediction Metrics

The prediction metrics are computed over agents. An agent is characterised only by its position in the ground plane and its forecast trajectory. The predicted agents for a frame are assigned to the ground truth (GT) agents of that frame by Hungarian matching on the centre distance, with the same allowed distance of 2 m as in Section 2.4.2. The track identity is never taken into account, so the assignment is recomputed from scratch in every frame rather than carrying established pairings as in tracking. The confidence of a prediction is likewise not read, so the counts are evaluated at the single operating point at which the tracker emitted its tracks.

A matched agent is additionally only useful if its forecast is close to what actually happened, which is what the following metrics quantify. For an agent with $K$ predicted future trajectories $\hat{y}^{(k)}$, a ground-truth future $y$ and the set $\mathcal{T}$ of future time steps with a valid annotation:

1. **minADE** is the average displacement of the best of the $K$ modes over the whole horizon, which rewards a model for containing one good hypothesis without punishing it for the diversity of the others,

$$\text{minADE}_K = \min_{k \in \{1,\dots,K\}} \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left\| \hat{y}_t^{(k)} - y_t \right\|_2. \tag{2.9}$$

A single annotated future step is sufficient for an agent to contribute, and the average is taken over $|\mathcal{T}|$ rather than over the full horizon, so agents that leave the scene early are scored on a correspondingly shorter forecast.

2. **minFDE** evaluates only the endpoint of the horizon, where the uncertainty of the forecast is largest,
$$\text{minFDE}_K = \min_{k\in\{1,\dots,K\}} \left\|\hat{y}_T^{(k)} - y_T\right\|_2. \tag{2.10}$$
It is defined only for agents that are still annotated at the end of the horizon, that is $T \in \mathcal{T}$. An agent that disappears earlier is excluded entirely
3. **Miss Rate (MR)** turns the endpoint error into a binary outcome and reports how often none of the K modes lands near the true endpoint,
$$\text{MR} = \frac{\left|\{\, i : \text{minFDE}_K^{(i)} > \tau \,\}\right|}{N_{\text{surv}}}, \qquad \tau = 2\,\text{m}, \tag{2.11}$$
where the index $i$ runs over the $N_{\text{surv}}$ matched agents that survive the whole horizon, since the others have no minFDE to threshold.
4. **End-to-end Prediction Accuracy (EPA)** [39] is the metric that judges the two tasks jointly. An agent contributes to the numerator only if it was both matched and forecast accurately, and every false positive is subtracted with a penalty $\alpha$,
$$\text{EPA} = \max\left(0,\ \frac{N_{\text{hit}} - \alpha\, N_{FP}}{N_{GT}}\right), \qquad \alpha = 0.5, \tag{2.12}$$
where $N_{\text{hit}}$ counts the matched agents whose $\text{minFDE}_K$ does not exceed $\tau$, and $N_{GT}$ the annotated agents that survive the whole horizon. Unlike the three metrics above, which are averaged over the agents that were found, EPA is normalised by the annotated agents rather than the matched ones, so *a model cannot improve it by detecting fewer objects and forecasting only the easy ones*.

The metrics are complementary and have to be read together. A model may reach a low minADE while missing many agents entirely, because the error is averaged only over the matched ones, and MR exposes the cases where no mode comes close at all. EPA penalises both failure modes at once and is therefore used as the primary criterion in this work.

# Chapter 3

# Related Work

TODO: add equations to important parts TODO: add livip3d repo and code mention this work is build upon it

Chapter 2 described the field as a whole. This chapter is narrower and covers only the works from which the model studied in this thesis is assembled. It begins with the query-based formulation that the model inherits, continues with the components introduced by the LiDAR branch, and closes with the two works that preceded this thesis in taking ViP3D as the base model and extending it with the LiDAR modality.

## 3.1 Query-Based Detection and Tracking

The architecture studied here descends from a line of work that removed many hand-designed components like non-maximum suppression (NMS) procedure or anchor generation. DETR [49] formulates detection as the direct prediction of a fixed-size set of object, in which a small number of learned queries attend to the image features and a Hungarian matching assigns every query to at most one annotated object.

The slow convergence caused by every query attending to every pixel was addressed by Deformable DETR [50], which lets a query sample only a few learned offsets around a reference point and thereby reduces both the cost and the number of epochs required for training. The reference point of a query is obtained by a linear projection of the positional part of its embedding, followed by a sigmoid that maps it into normalised coordinates. In the same work the authors introduce a two-stage variant that initialises the reference points from the centres of the highest scoring region proposals generated on top of the encoder output, so that the decoder starts from positions determined by the input rather than by a learned constant, and this variant outperforms the single-

stage one [50]. DETR3D carried the formulation into three dimensions by placing the reference point of each query in 3D space and projecting it into the available camera views, so that multi-view images are consumed without an intermediate dense depth estimate [33].

The step from detection to tracking was taken by MOTR, which keeps a query alive across frames instead of detecting the same object independently in each one. The identity of an object is then carried by the query itself rather than recovered afterwards by an association heuristic [51]. How reliably that identity survives across frames is what the tracking metrics of Section 2.4.2 measure. ViP3D combines the two ideas and extends them to forecasting. Agent queries are initialised in the DETR3D manner, propagated through time in the MOTR manner by a memory bank and a query interaction module, and read directly by the trajectory decoder. The result is a pipeline that is differentiable from the camera images to the predicted trajectories and that contains no thresholded intermediate representation [39]. ViP3D is the model this thesis builds upon.

## 3.2 LiDAR Backbones and Heatmap Query Initialisation

TODO: add other lidar backbones

ViP3D perceives the scene through cameras alone, so its queries are initialised from a fixed learned embedding that is identical in every frame regardless of what the scene contains. Replacing that embedding requires a representation of the point cloud and a way of reading object hypotheses out of it.

The representation follows the established encoders described in Section 2.1.2.3. PointPillars groups the points into vertical columns and flattens them into a pseudo-image, which allows a conventional two-dimensional backbone to process the point cloud at a cost that does not depend on its density [25], and SECOND forms the BEV feature map with sparse convolutions that skip the empty regions of the grid [26].

The hypotheses come from CenterPoint, which abandons anchors and instead predicts a class-wise heatmap over the BEV grid, where every peak marks the centre of an object [32]. TransFusion [52] turned this into the initialisation of a transformer decoder. *The highest scoring heatmap peaks become object queries whose content is sampled from the BEV feature* at the peak and whose reference point is the metric position of that cell, so that the decoder begins from a hypothesis that depends on the input rather than from a constant embedding. *This is the mechanism that replaces the learned query embedding of ViP3D in the present work.*

## 3.3 Attention-Based Camera-LiDAR Fusion

After the queries originate from the point cloud 3.2, the camera modality may be introduced to supply the colour and texture that the point cloud does not contain. TransFusion [52] performs this both in the initialization procedure and by adding a second decoder layer that cross-attends to the image features by projecting the queries' location into image plane like [50]. It trains that layer on top of a frozen LiDAR model, which prevents the fusion from destabilising a branch that was already pretrained beforehand. The attention in that layer is spatially modulated following SMCA [53], which adds the logarithm of a Gaussian centred on the projection of the reference point of the query to the attention logits before the softmax, so that a query attends only where it can geometrically be and converges considerably faster than unconstrained cross-attention.

The initialisation itself operates in the BEV plane rather than in the image. TransFusion collapses the features of each image along the vertical axis, so that one image column becomes a single token, on the observation that a BEV location maps to a column by camera geometry and that a column rarely holds more than one object. The BEV cells then attend to those columns, the result is averaged over the cameras, and a second heatmap predicted from it is combined with the LiDAR one, so that the camera takes part in deciding which queries are created at all.

## 3.4 Trajectory Decoding from Agent Features

The last stage turns the state of an agent into a set of possible futures. VectorNet replaced the rasterised scene encodings common at the time with a vectorised one, in which agent histories and map elements are represented as polylines and a hierarchical graph aggregates first within and then across them [34]. LaneGCN refined the treatment of the map by building an explicit lane graph and exchanging information between the agents and the lanes in both directions [35]. The decoder of ViP3D follows this line, with one substantial difference. It does not receive an observed trajectory assembled by a tracker, but the feature vector of the agent query itself, which still carries the visual evidence that a bounding box would have discarded [39].

## 3.5 Positioning of This Work

The idea of giving ViP3D a LiDAR branch has been pursued before in the group in which this thesis was written. Li-ViP3D [54] addresses limited capability of exploitation of the complementarity between camera and LiDAR data in the current perception and prediction (PnP) solutions by extending ViP3D query-based camera-only pipeline with

camera-LiDAR fusion and reports improved end-to-end perception and prediction on nuScenes. Li-ViP3D++ [55] develops the fusion further by gating the contribution of the camera on the query and by sampling the image features in deformable manner. Both of these works treat the addition of LiDAR as a fusion problem, in which the two modalities are combined for queries that are still generated the way ViP3D generates them from a learned embedding that never depends on any particular scene.

The present work asks where the queries should come from and tackles the problem of their initialisation. As mentioned in 3.1, Deformable DETR [50] indroduced a two-stage variant initialises the reference points from the centres of the highest scoring region proposals generated on top of the encoder output, so that the decoder starts from positions determined by the input. This variant outperformed the single-stage one, which is evidence that reference points derived from the data are prefereable to reference points learned as constants. In this thesis, the same principle was applied in the BEV plane. The proposal generator is a class-wise heatmap predicted from the LiDAR feature map, and its highest scoring peaks supply the query with both the reference point from the metric position of the peak and the content from the feature sampled at that cell.

This work is therefore positioned differently from the two that precede it. Li-ViP3D and Li-ViP3D++ combine the two modalities for queries that are still created the way ViP3D creates them, so the LiDAR enters at the moment a query is refined. Here it enters earlier, at the moment a query is created, and the camera is reintroduced only afterwards in a second decoder layer trained on top of the frozen LiDAR model. Fusion is thus a refinement of a hypothesis that the point cloud has already formed.

## 3.6 Summary and Starting Points

TODO: Summary of Chapters 2 and 3 ... several paragraphs... 1/2 to 1 page

Based on the study presented in Chapters 2 and 3, the following gaps are identified with potential to improve:

- Query-based PnP models still create their queries from a learned embedding that is identical in every frame, although the detection literature has already shown that reference points derived from the input outperform reference points learned as constants.
- The works that add LiDAR to ViP3D use it to refine queries rather than to create them, so it is not known what the query-based formulation achieves when the queriese themselves originate in the point cloud.
- A query that already begins at a position read from the point cloud has little left to correct, which should allow a far shallower decoder. ViP3D moves its queries

towards the objects through several refinement layers, whereas an initialisation of this kind leaves only the camera layer to refine. The latency this saves and the accuracy of predictions it costs have not been measured.
- The contribution of the camera has been measured only on top of camera-initialised models, so it remains unclear which metrics and which object classes still benefit once the queries already carry geometric evidence.

The goal of this thesis is therefore to develop and evaluate a deep learning architecture for end-to-end perception and prediction in which the agent queries originate in the point cloud, extending ViP3D with a LiDAR branch in place of its learned query embedding.

To fill in these gaps, the following starting points are proposed.
- Starting point 1: replace the learned query embedding of ViP3D with a class-wise heatmap predicted from the LiDAR BEV feature map, taking the reference point of a query from the metric position of a peak and its content from the feature sampled at that cell.
- Starting point 2: reintroduce the camera as a single refinement layer with spatially modulated cross-attention, trained on top of the frozen LiDAR model, and ablate it so that the effect of the fusion can be evaluated separately.
- Starting point 3: reduce the decoder to one layer reading the BEV map and that one camera layer, and report the metrics against measured latency so that the saving attributable to the initialisation is separated from the cost of less refinement.

**Priebežná správa BP1/DP1 musí obsahovať:**

1. Prehľad podobných riešení alebo produktov, s ktorými sa študent oboznámil pri štúdiu danej problematiky.
2. Špecifikáciu požiadaviek na vytvárané riešenie alebo produkt (Chapter 4).
3. Návrh
4. Vlastne zhodnotenie práce za príslušné obdobie (Chapter C - Section C.1).

# Chapter 4

# Objectives and Methodology

The present work applies the same principle in the BEV plane. The proposal generator is a class-wise heatmap predicted from the LiDAR feature map, and its highest scoring peaks supply both halves of a query, the reference point from the metric position of the peak and the content from the feature sampled at that cell together with an embedding of its class. Only the slots left free by the tracker are filled in this way, so that agents already being followed keep the queries they carry and the initialisation concerns newly appearing objects alone. The queries entering the decoder are therefore hypotheses about the scene currently observed rather than constants inherited from training, and the camera is reintroduced only afterwards, in a second decoder layer trained on top of the frozen LiDAR model, which makes the fusion a refinement of those hypotheses rather than their source.

## 4.1 Objectives

Vybrané ciele pre navrhnuté riešenie alebo špecifikácia požiadaviek (funkčné a nefunkčné požiadavky) na naimplementovaný softvérový produkt.

- povinne pre DP a DiZP, ale aj BP môže
- max 2 pages for the whole chapter.

As presented in Chapter 2, the area of... is wide with many interesting unsolved problems. Therefore, the goals of this thesis are focused on selected objectives (maximum 3 objective) to... as follows:

- Objective 1:
- Objective 2:
- Objective 3:

## 4.2 Methodology

The following steps have been chosen for the design and development of ...

- Step 1:
- Step 2:
- ...
- Step n:

These steps are commitments to each other in the context of ... of the work.

# Chapter 5

# Design / Methods Description

Concrete header for the proposed design of the thesis's solution TODO: Hungarian matching

This chapter can contain:

- Specification based on the Sections 3.6
- Design
- Technologies (can also be/moved in/to Chapter 6 Implementation)

Description how the design works (as much as possible from the below list)

- We need to separate our design from our concrete implementation
- Architecture, Block scheme, Event diagram, Sequence diagram, Algorithms, ...
- Text description of each blocks and their connections and/or dependencies
- Time complexity and memory complexity estimation for each execution step and the whole design. Communication complexity of the design (if any).

Figure 5.1 is about ... description of each blocks

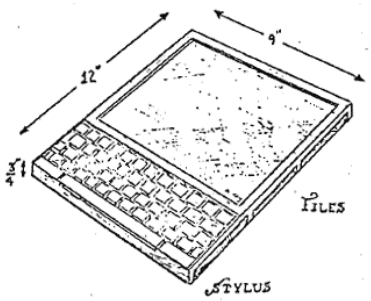



Figure 5.1: Architecture/Workflow

# Chapter 6

# Implementation

Concrete header for the implementation of the proposed design

The concrete realization of our proposed design

Concrete tools, software, solutions based on concrete dataset

Pseudocode of Algorithms, or Code snippets

$$\text{Bounce Rate} = \frac{\text{Single page sessions}}{\text{Total sessions}} \tag{6.1}$$

# Chapter 7

# Experiments and Evaluation

## 7.1 Setup

hardware,
software, libraries, versioning,
virtual environment description and setting,
datasets, NDA,
etc.

| Machine | Remote | Google Colab | Kaggle | FIIT Cloud | Local |
|---|---|---|---|---|---|
| Python | 3.9.19 | 3.10.12 | - | - | 3.10.12 |
| PyTorch | - | - | - | - | - |
| TensorFlow | 2.10 | 2.15 | - | - | 2.14 |
| GPU (NVIDIA) | RTX xx | Tesla T4 15 GB | Tesla P100 16GB | - | - |
| CUDA | 12.2 | 12.2 | - | - | - |
| cuDNN | 8.1.0 | N/A | - | - | - |
| CUDA Toolkit | 11.2.2 | N/A | - | - | - |
| RAM | 32/64 GB | 51 GB | 30 GB | 32 GB | 16 GB |
| CPU | 8x cores Intel i7 | - | Xeon 4x cores | 4x cores | - |
| Storage | SSD | Google Driver | 73 GB | 128 GB | HDD |

Table 7.1: Hardware, software and versioning used for realization

## 7.2 Case Study - Experiment 01

## 7.3 Case Study - Experiment 02

## 7.4 Case Study - Experiment 03

## 7.5 Summary

Main *key findings* of the work based on the Experiment results

- add ...
- ...

*Discussion* about various considerations based on the experiment results (if needed).

Regarding *limitations* of the work ...

- add ...
- ...

# Chapter 8

# Conclusion

## 8.1 Summary

Summary of the whole work including SOTA, design, implementations, experiments, evaluations, key fingdings ... 1 page

## 8.2 Future Work

1/2 page

# Resumé

Chapter je bez cislovania, pouziva sa len **boldovanie textu** na zyraznenie useku textu. Bez Figure, Table, atd. ak treba tak odvolava sa na originalne Figure a Table.

10% v slovenčine pre tých, ktorí majú prácu v EN: preklad

| EN | SK |
|---|---|
| Introduction | primerane strucne |
| SOTA | primerane strucne |
| SOTA - cast Summary and Starting Points | cele |
| Objective and Methodology | cele |
| Design, Implementation | primerane strucne |
| Experiments and Evaluation | primerane strucne |
| Experiments and Evaluation - cast Summary | cele |
| Conclusion and Future Work | cele |

Table 8.1: Resumé

# List of Abbreviations

**ADE** Average Displacement Error

**AMOTA** Average Multi-Object Tracking Accuracy

**AMOTP** Average Multi-Object Tracking Precision

**AP** Average Precision

**AV** autonomous vehicle

**AVs** autonomous vehicles

**BEV** bird's-eye view

**CNN** convolutional neural network

**DBN** dynamic Bayesian network

**EPA** End-to-end Prediction Accuracy

**FDE** Final Displacement Error

**GMM** Gaussian mixture model

**GNN** graph neural network

**GP** Gaussian process

**GPS** global positioning system

**GT** ground truth

**HD map** high-definition map

**HMM** hidden Markov model

**IDS** identity switch

**IL** imitation learning

**IMU** inertial measurement unit

**IoU** intersection over union

**IRL** inverse reinforcement learning

**LiDAR** light detection and ranging

**LSTM** long short-term memory

**mAP** mean Average Precision

**MOTA** Multi-Object Tracking Accuracy

**MOTP** Multi-Object Tracking Precision

**MR** Miss Rate

**NDS** nuScenes Detection Score

**NMS** non-maximum suppression

**PnP** perception and prediction

**RL** reinforcement learning

**RNN** recurrent neural network

**SLAM** simultaneous localization and mapping

**TCN** temporal convolutional network

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# Appendix A

# Description of Digital Submission

**Thesis Evidence Number:**
FIIT-XXXX-XXXXX

Obsah elektronického média je väčší ako 1GB a preto je odovzdaný vedúcemu záverečnej práce (doc. Ing. Giang Nguyen Thu, PhD.).

**Name of submitted archive:** `DP_MenoPriezvisko.zip`

The digital part included in the thesis contains the following files:

- Ubuntu OS: run in command line `$ tree` to get a directory structure as below.
- Inspiration from *https://github.com/FIIT-ISA/cookiecutter-data-science*

```
ROOT // Root directory.
│
├── flatbuffers // This flatbuffers director
└── scheme
```

```
├── LICENSE
├── Makefile           <- Makefile with commands like `make data` or `make train`
├── README.md          <- The top-level README for developers using this project.
├── data
│   ├── external       <- Data from third party sources.
│   ├── interim        <- Intermediate data that has been transformed.
│   ├── processed      <- The final, canonical data sets for modeling.
│   └── raw            <- The original, immutable data dump.
│
├── docs               <- A default Sphinx project; see sphinx-doc.org for details
│
├── models             <- Trained and serialized models, model predictions, or model summaries
│
├── notebooks          <- Jupyter notebooks. Naming convention is a number (for ordering),
│                         the creator's initials, and a short `-` delimited description, e.g.
│                         `1.0-jqp-initial-data-exploration`.
│
├── references         <- Data dictionaries, manuals, and all other explanatory materials.
│
├── reports            <- Generated analysis as HTML, PDF, LaTeX, etc.
│   └── figures        <- Generated graphics and figures to be used in reporting
│
├── requirements.txt   <- The requirements file for reproducing the analysis environment, e.g.
│                         generated with `pip freeze > requirements.txt`
│
├── setup.py           <- makes project pip installable (pip install -e .) so src can be impor
├── src                <- Source code for use in this project.
│   ├── __init__.py    <- Makes src a Python module
│   │
│   ├── data           <- Scripts to download or generate data
│   │   └── make_dataset.py
│   │
│   ├── features       <- Scripts to turn raw data into features for modeling
│   │   └── build_features.py
│   │
│   ├── models         <- Scripts to train models and then use trained models to make
│   │   │                 predictions
│   │   ├── predict_model.py
│   │   └── train_model.py
│   │
│   └── visualization  <- Scripts to create exploratory and results oriented visualizations
```

Figure A.1: The structure of the ZIP file

# Appendix B

# Technical Documentation

## B.1 Installation Manual

1. Clone the project from the Git repository (URL) OR unzip the ZIP file ...
2. Run *pip install -r requirements.txt* in the root folder of the project
3. Run *python -m spacy download en* to have the data corpus ...

## B.2 User Manual / Používateľská príručka

1. Spustenie programu funguje pomocou hlavného Python súboru *main.py* príkazom *python main.py*.
2. Po spustení programu sa zobrazí grafické rozhranie programu v ktorom sa dá hneď na hlavnej obrazovke spustiť sumarizácia pre jeden vstupný súbor.
3. Tlačidlami v hornej časti obrazovky sa dá presúvať medzi funkcionalitami programu ako je viac súborová sumarizácia alebo zobrazovanie článkov. Na týchto obrazovkách sa tlačidlom *Back to main menu* dá vrátiť späť na hlavnú obrazovku.

Screenshot(s) používateľského grafického rozhrania (GUI) ako obrázok (Figure)

(Another variant without GUI: ) The main objective of this work is to ... Several case studies were presented that examine the capabilities of ... to show the feasibility and achievability of ... Users in our work are considered to work directly with the code as described in Section B.1.

## B.3 Developer Manual

if needed then add ...

# Appendix C

# Work Schedule

## C.1 Work Schedule for BP1/DP1

| Week | Plan |
|---|---|
| 1. - 2. | ... |
| 3. - 4. | ... |
| 5. - 6. | ... |
| 7. - 8. | ... |
| 9. - 10. | ... |
| 11. - 12. | ... |

Some explanation text... (if needed) ... a few sentences

**Self-Evaluation of Work Schedule for BP1/DP1:** Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 1 work plan is considered satisfied and successful.

## C.2 Work Schedule for BP2/DP2

| Week | Plan |
|---|---|
| 1. - 2. | ... |
| 3. - 4. | ... |
| 5. - 6. | ... |
| 7. - 8. | ... |
| 9. - 10. | ... |
| 11. - 12. | ... |

Some explanation text... (if needed) ... a few sentences

**Self-Evaluation of Work Schedule for BP2/DP2:** Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 2 work plan is considered satisfied and successful.

## C.3 Work Schedule for DP3

| Week | Plan |
|---|---|
| 1. - 2. | ... |
| 3. - 4. | ... |
| 5. - 6. | ... |
| 7. - 8. | ... |
| 9. - 10. | ... |
| 11. - 12. | ... |

Some explanation text... (if needed) ... a few sentences

**Self-Evaluation of Work Schedule for DP3:** Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The DP 3 work plan is considered satisfied and successful.